

Principles of Information Security

Mid Semester Examination Question Bank

1. Define the length-preserving, keyed function F as follows -

$$F_k(x) = k + x$$

Prove that F is a pseudo random function by describing and analyzing the concrete distinguished.

2. Suppose there is only one public channel between Alice and Bob and it has been given that no "active" adversary (there may be passive adversaries) is "active" over this channel. Alice wants to send a message securely to Bob at time t so that Bob can get the message after time $t + t_1$, where t_1 can be significantly large ($t_1 \gg t$). Assuming that the channel will have no transmission delay, can you create a simple protocol for secure communication between Alice and Bob using the concepts of Private and Public key cryptography?
3. Define a Hash Function and the properties that it must possess from a cryptographic point of view. Explain the Birthday Paradox and the Birthday Attack in brief.
4. Let F be a pseudorandom function. Show that the following MAC for messages of length $2n$ is insecure: Gen outputs a uniform $k \in \{0, 1\}^n$. To authenticate a message $m_1 || m_2$ with $|m_1| = |m_2| = n$, compute the tag $F_k(m_1) || F_k(m_2)$.
5. Let F be a pseudorandom function. Show that each of the following MACs is insecure, even if used to authenticate fixed-length messages. (In each case Gen outputs a uniform $k \in \{0, 1\}^n$. Let $\langle i \rangle$ denote an $n/2$ -bit encoding of the integer i .)
 - (a) To authenticate a message $m = m_1, \dots, m_\ell$, where $m_i \in \{0, 1\}^n$, compute
$$t := F_k(m_1) \oplus \dots \oplus F_k(m_\ell).$$
 - (b) To authenticate a message $m = m_1, \dots, m_\ell$, where $m_i \in \{0, 1\}^{n/2}$, compute
$$t := F_k(\langle 1 \rangle || m_1) \oplus \dots \oplus F_k(\langle \ell \rangle || m_\ell).$$

- (c) To authenticate a message $m = m_1, \dots, m_\ell$, where $m_i \in \{0, 1\}^{n/2}$, choose uniform $r \leftarrow \{0, 1\}^n$, compute

$$t := F_k(r) \oplus F_k(\langle 1 \rangle \| m_1) \oplus \dots \oplus F_k(\langle \ell \rangle \| m_\ell),$$

and let the tag be $\langle r, t \rangle$.

6. Consider a variant of CBC-MAC where the message size is appended to the end of the message prior to passing to basic CBC-MAC. Prove that for any arbitrary-length messages, this variant is not secure for
7. Consider a MAC for arbitrary-length messages by

$$\text{Mac}_{s,k}(m) = H^s(k \| m)$$

where H is a collision-resistant hash function. Is this a secure MAC when H is constructed via the Merkle-Damgård transform? (Assume the hash key s is known to the attacker, and only k is kept secret.)

8. Show how to find a collision in the Merkle tree construction if t is not fixed. Specifically, show how to find two sets of inputs $x_1, \dots, x_t$ and $x'_1, \dots, x'_{2t}$ such that $\mathcal{MT}_t(x_1, \dots, x_t) = \mathcal{MT}_{2t}(x'_1, \dots, x'_{2t})$
9. Consider the modified Merkle-Damgård transform: The input length is not included at all (i.e., output z_B and not $z_{B+1} = h^s(z_B \| L)$). (Assume the resulting hash function is only defined for inputs whose length is an integer multiple of the block length.) Determine whether the result is collision resistant. If yes, provide a proof; if not, demonstrate an attack.

Exercise 1 Consider a one-time pad encryption which instead of the binary alphabet, uses the English alphabet and is defined for messages of length 5 (i.e. each message is a 5 letter string). For example, if $\text{Gen}(1^n)$ returns a string like $\text{sk} = \text{'AQDCR'}$ and the message is $m = \text{'BBBBB'}$, then m is encrypted as a ciphertext $c = \text{'BREDs'} = \text{Enc}_{\text{sk}}(m)$ (i.e. we treat letters 'A', 'B', 'C' as 0, 1, 2... and perform addition modulo 26). Call this scheme $S = (\text{Gen}, \text{Enc}, \text{Dec})$.

Now since S is a one-time pad, Gen returns every 5 letter string with equal probability—including the string $\text{sk} = \text{'AAAA'}$.

1. If the message $m = \text{'HELLO'}$ and $\text{sk} = \text{'AAAA'}$, what is the cipher text?
2. Consider a scheme $S' = (\text{Gen}', \text{Enc}, \text{Dec})$ where Gen' uniformly samples an element from the set of all 5 letter strings, excluding the string 'AAAA'. Is S' as secure as S ? If yes, prove that S' satisfies semantic security. If not, prove that S' does not satisfy semantic security.

Exercise 2 The One-Time Pad

1. Prove that

$$\Pr[M = m | C = c] = \Pr[M = m]$$

is equivalent to

$$\Pr[C = c | M = m_0] = \Pr[C = c | M = m_1]. \quad (1)$$

2. Suppose the One-Time Pad is used as a 'Two-Time Pad', i.e. suppose the same key is used to encrypt two different messages m_A and m_B . Suppose it is known that m_A and m_B are independently uniformly sampled to be 0 or 1. Can Equation (1) hold in this case? Prove it either way.

Exercise 3 Towards computational security.

CONSTRUCTION 3.17

Let G be a pseudorandom generator with expansion factor l . Define a private-key encryption scheme for messages of length l as follows:

- **Gen**: on input 1^n , choose uniform $k \in \{0, 1\}^n$ and output it as the key.
- **Enc**: on input a key $k \in \{0, 1\}^n$ and a message $m \in \{0, 1\}^{l(n)}$, output the ciphertext $c := G(k) \oplus m$.
- **Dec**: on input a key $k \in \{0, 1\}^n$ and a ciphertext $c \in \{0, 1\}^{l(n)}$, output the message $m := G(k) \oplus c$.

A private-key encryption scheme based on any pseudorandom generator.

Figure 1: Encryption scheme using a PRG. Taken from [KL14]

1. Can any encryption scheme on message space $\mathcal{M}$ be secure (against unbounded adversaries) if the key space $\mathcal{K}$ is smaller than the message space, i.e. if $|\mathcal{K}| < |\mathcal{M}|$? See Theorem 2.10 from [KL14] if you get stuck—but this should be quite straightforward to prove.
2. What are the two relaxations to our security requirement we started with for the one-time pad, to get to computational security? Define any new terms you use.
3. Recall the three security games formalising security against computationally bounded adversaries.
 - (a) Single ciphertext security, equivalently semantic security (IND; §3.2 [KL14])
 - (b) Chosen Plaintext Security (IND-CPA; §3.4.2 [KL14])
 - (c) Chosen Ciphertext Security (IND-CCA; §3.7.1 [KL14])

Briefly justify why each of these notions capture successively stronger notions of security.

Exercise 4 (IND secure encryption). Consider the encryption scheme S in Figure 1.

1. Define a PRG with expansion factor 1.
2. Prove that if a G is a secure pseudorandom generator, then the scheme S satisfies semantic security.

Exercise 5 CPA secure encryption

1. Define a pseudorandom function? Again, what is the intuition behind the definition?
2. Prove that if F is satisfied the definition of a pseudorandom function, then the scheme 3.30 satisfies CPA security.
3. Suppose Enc samples the same r when invoked separately to encrypt messages m_1 and m_2 . What does this observation say about the security of the scheme?

Exercise 6. CPA and CCA Security

1. Consider an encryption scheme $S := (\text{Gen}, \text{Enc}, \text{Dec})$ where $\text{Gen}(1^\lambda)$ returns a random key sk but given the secret key sk , Enc_{sk} is a deterministic function. Suppose someone claims S is a CPA secure encryption scheme. Do you need more information to determine if their claim is valid? If not, why not?

CONSTRUCTION 3.30

Let F be a pseudorandom function. Define a private-key encryption scheme for messages of length n as follows:

- **Gen**: on input 1^n , choose uniform $k \in \{0,1\}^n$ and output it.
- **Enc**: on input a key $k \in \{0,1\}^n$ and a message $m \in \{0,1\}^n$, choose uniform $r \in \{0,1\}^n$ and output the ciphertext

$$c := (r, F_k(r) \oplus m).$$

- **Dec**: on input a key $k \in \{0,1\}^n$ and a ciphertext $c = (r, s)$, output the plaintext message

$$m := F_k(r) \oplus s.$$

A CPA-secure encryption scheme from any pseudorandom function.

Figure 2: CPA secure encryption scheme using

2. Consider a variant S' of the scheme S in figure 2: S' is the same as S except that S' uses Enc' . The scheme S' is designed to be used as follows:
 - (a) A single party uses Enc' and this party keeps a counter r' that is incremented every time a message is encrypted (starting from $r' = 0$).
 - (b) Enc' is the same as Enc except that r' is used instead of r .
 - (c) In the CPA security definition, assume that the 'encryption oracle' essentially asks this party to produce encryptions.

Do you think this scheme is CPA secure, given the constraints above? Give a semi-formal proof of your conclusion. Explain how your answer is consistent with your answer to question 1 above.

3. Recall that in Exercise 2, question 3, we informally discussed what it means for an encryption scheme to be homomorphic. Suppose an encryption scheme satisfies the homomorphic property for $*$ where $*$ is an operation satisfying the following: there exists m_{zero} such that $m * m_{\text{zero}} = m$ for all messages m . Do you see any reason why such a scheme cannot satisfy CPA security? How about CCA security? If you claim that security is impossible in either case, prove it.

Exercise 7 OWFs

1. Formally define the following
 - (a) One-way function
 - (b) Hard-core predicate

2. Is it known that for every one-way function, one can construct a hard-core predicate? What is the Goldreich-Levin theorem (given in book)?
3. Give a construction for the following,
 - (a) a PRG from a OWF;
 - (b) a PRF from a PRG;
4. Is a PRF not just a PRG with an expansion factor of $n \cdot 2^n$? Why do we need a separate construction for PRFs then?
5. Show that
 - (a) a PRG implies a OWF exists.
 - (b) an IND secure encryption scheme (where messages are twice as long as the key) implies a OWF exists.